



江苏省苏州中学国际部
Suzhou High School – International Division

SZZX-ID BP Physics

Practical: Properties of an electromagnet

Class Outline

Title of Unit	Introduction to magnetic field
Title of Lesson	Practical: Properties of an electromagnet
Starter Activity	Show a sample hand-made electromagnet
Direct Instructions	Teaching Strategies: Demonstration, Lecture, Collaboration, Questioning, Inquiry
	- Inquiry: practical
	Differentiation: Content, Instructional Strategies, the Classroom, Products, Teacher
	- Lab work in pairs
	International Mindedness:
Lesson Objectives	By the end of the lesson, students will be able to: • Make a simple electromagnet • Verify the right-hand grip rule with the electromagnet • Determine how increasing the voltage supply affects the strength of an electromagnet • Determine how increasing the number of coils affects the strength of an electromagnet
ATLs	Identify unit ATLs and how they are being addressed in the lesson
	Research skill
IB Learner Profile	Link the lesson to the characteristics of the IB Learner attribute
	Inquirer, thinker
Teaching Activities	- Structure of an electromagnet - Safety precaution: prevent overheating of the battery
Plenary Session	Checking for understanding / Homework / Q & A / Reflection etc.
	Worksheet submission requirement

Objective of experiment



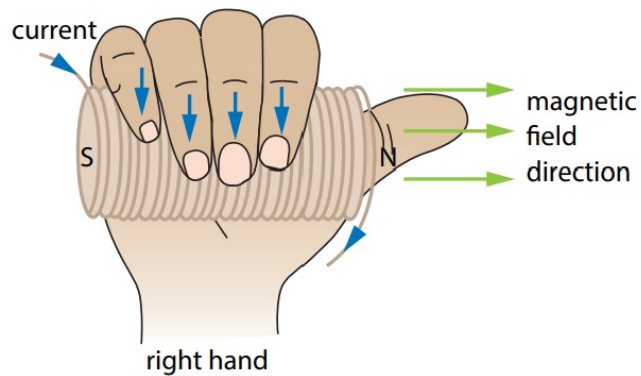
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- Make a simple electromagnet
- Verify the right-hand grip rule with the electromagnet
- Determine how increasing the voltage supply affects the strength of an electromagnet
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Revision - Right Hand Grip Rule



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The direction of the magnetic field around a solenoid can be found using the “right-hand grip rule”, too

- Use right hand
- The fingers curl in the direction of the current
- The thumb points in the direction of the magnetic field

Safety precaution



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To enhance the magnetic effect, a high current runs through the wire and batteries.

To avoid overheating of the wire and battery, disconnect the wire as soon as possible.